SOIYOD SIFAT MAHMUD

Electronics and Telecommunication Engineering
Chittagong University of Engineering and Technology (CUET)
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PROFESSIONAL SUMMARY

Final-year Electronics and Telecommunication Engineering student with hands-on experience across software development, artificial intelligence, embedded systems, IoT, digital electronics, VLSI design, and computer architecture. Experienced in developing AI-powered applications, ESP32-based IoT systems, autonomous robotics, and full-stack web applications. Strong programming background in Python, C/C++, and JavaScript, with practical knowledge of VLSI design, Cadence Virtuoso, SAP-1 computer architecture, digital logic design, networking, and cybersecurity. Interested in AI/ML, software engineering, embedded systems, VLSI, IoT, and telecommunication technologies.

EDUCATION

B.Sc. in Electronics and Telecommunication Engineering (ETE) **2026**
Chittagong University of Engineering and Technology (CUET), Bangladesh
Current CGPA: 3.65/4.00

TECHNICAL SKILLS

Programming: Python, C, C++, JavaScript, HTML, CSS
AI/ML: Machine Learning, Deep Learning, Computer Vision, NLP, Speech Emotion Recognition, OpenCV, YOLO, MobileNet SSD
Web Development: React, Vite, Tailwind CSS, FastAPI, Node.js, Express.js, REST API
Database: SQLite, MongoDB, ChromaDB, Google Sheets
Embedded/IoT: ESP32, Arduino, Raspberry Pi, MQTT, Sensors, Arduino IDE, Real-Time Monitoring
VLSI/Digital Design: VLSI Design, Cadence Virtuoso, SAP-1 Computer Architecture, Logisim
Networking/Security: Cisco Networking, Wireshark, Nmap, Packet Tracer, Cybersecurity Fundamentals
Tools: Git, GitHub, MATLAB, Flask, Tesseract OCR

EXPERIENCE / INDUSTRIAL TRAINING

Industrial Training – FactoryNext

- Worked on sensor-based and ESP32 projects, gaining hands-on experience in IoT, embedded systems, real-time monitoring, and microcontroller-based development.
- Built a **Smart Temperature & Humidity Monitoring System** using ESP32, Arduino IDE, MQTT, database integration, and a custom web dashboard for real-time monitoring and visualization.

Industrial Attachment – Brain Station 23

- Studied fintech applications in Bangladesh, including internet and mobile banking, BQR, digital payments, investment schemes, and financial products.
- Explored practical applications of AI in financial technology and developed an AI-powered Investment Robo-Advisor Chatbot tailored to the Bangladeshi market.

PROJECTS

Mentora – AI-Powered PDF Learning Platform

- Developed an AI-powered platform for extracting and analyzing text, images, tables, diagrams, charts, equations, and handwritten content from PDF documents.
- Built with React, Vite, Tailwind CSS, FastAPI, PyMuPDF, Tesseract OCR, ChromaDB, SQLite, and LLM APIs.

VisionDrive – Autonomous Monitoring Robot

- Developed an autonomous robot combining embedded systems, robotics, and edge AI for real-time object detection and counting.
- Used Arduino UNO for motor control and Raspberry Pi with MobileNet SSD and Flask MJPEG for real-time computer vision.

ProgressNest – Full-Stack Personal Growth Platform

- Built a full-stack platform for tracking personal goals, achievements, productivity, and collaborative learning activities.

Smart Home Safety & Security System

- Developed a real-time safety system integrating smoke and gas sensors, LDR, laser-based intrusion detection, person counting, Arduino control, and buzzer alerts.

CERTIFICATIONS

- **Engineering Design and Analyses with MATLAB** – Bangladesh Computer Council
- **MATLAB Certified** – MATLAB
- **Advent of Cyber** – TryHackMe
- **Introduction to Cybersecurity** – Cisco Networking Academy
- **Networking Basics** – Cisco Networking Academy